15.095 Machine Learning HW1

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Question 1

(a)

False.

First, the primal is infeasible. The first two constraints imply

$$x \leq y$$

and

$$-x + y \leq -1 \quad \Longrightarrow \quad y \leq x - 1.$$

Any feasible solution would have to satisfy

$$x \leq y \leq x - 1,$$

which is impossible. Hence, the primal is infeasible.

To determine whether the dual is unbounded, we can rewrite the primal constraints as

$$x - y \leq 0,$$

$$-x + y \leq -1.$$

Using the dual variables $u_1, u_2 \geq 0$ and the equation, the dual is

$$\min \quad -u_2$$

subject to

$$u_1 - u_2 \geq 1,$$

$$-u_1 + u_2 \geq 2,$$

$$u_1, u_2 \geq 0.$$

If both constraints must be true, their sum must also be true.

$$(u_1 - u_2) + (-u_1 + u_2) \geq 1 + 2,$$

so

$$0 \geq 3,$$

which is impossible. So, the dual is **infeasible**, not unbounded. The statement is false. □

(b)

False.

From lecture, for a Frobenius norm uncertainty set,

$$\mathcal{U}_{F(q)} = \left\{\Delta \in \mathbb{R}^{n\times p} : \|\Delta\|_{q-F} \leq \rho\right\},$$

where

$$\|\Delta\|_{q-F} = \left(\sum_{i,j} |\Delta_{ij}|^q\right)^{1/q},$$

we get the equivalence of robustness and regularization as

$$\min_{\beta} \max_{\Delta\in\mathcal{U}_{F(q)}} \|y - (X+\Delta)\beta\|_q = \min_{\beta} \|y - X\beta\|_q + \rho\|\beta\|_{q^*},$$

where

$$\frac{1}{q} + \frac{1}{q^*} = 1.$$

We take $q = 2$, and we have $q^* = 2$, so

$$\min_{\beta} \max_{\Delta:\|\Delta\|_{2-F}\leq\mu} \|y - (X+\Delta)\beta\|_2 = \min_{\beta} \|y - X\beta\|_2 + \mu\|\beta\|_2.$$

$$\|\Delta\|_{2-F} = \left(\sum_{i,j} \Delta_{ij}^2\right)^{1/2},$$

Since this places one bound on the perturbation across the **entire** matrix X, this corresponds to a *global* perturbation. Therefore,

$$\mu\|\beta\|_2 \quad \text{is a global perturbation bounded by } \mu.$$

For an induced norm uncertainty set,

$$\mathcal{U}_{(r,q)} = \left\{\Delta \in \mathbb{R}^{n\times p} : \|\Delta\|_{r,q} \leq \rho\right\},$$

$$\|\Delta\|_{r,q} = \max_{x} \frac{\|\Delta x\|_q}{\|x\|_r},$$

we have the theorem

$$\min_{\beta} \max_{\Delta\in\mathcal{U}_{(r,q)}} \|y - (X+\Delta)\beta\|_q = \min_{\beta} \|y - X\beta\|_q + \rho\|\beta\|_r.$$

Let $r = 1$, $q = 2$, and $\rho = \lambda$,

$$\min_{\beta} \max_{\Delta:\|\Delta\|_{1,2}\leq\lambda} \|y - (X+\Delta)\beta\|_2 = \min_{\beta} \|y - X\beta\|_2 + \lambda\|\beta\|_1.$$

The induced $(1,2)$-norm uncertainty set matches with *feature-wise* perturbations. Therefore, $\lambda\|\beta\|_1$ corresponds to feature-wise perturbations bounded by λ.

So, for the Elastic Net objective

$$\min_{\beta} \|y - X\beta\|_2 + \lambda\|\beta\|_1 + \mu\|\beta\|_2,$$

λ corresponds to feature-wise perturbations, and μ corresponds to global perturbations. □

(c)

True.

1. The objective

$$\min_{x,y} \|x - y\|_1 = \min_{x,y} \sum_{i=1}^{n} |x_i - y_i|.$$

Introduce variables $t_i \geq 0$,

$$\boxed{t_i \geq x_i - y_i, \quad t_i \geq -(x_i - y_i), \quad i = 1, \ldots, n.}$$

Since we are minimizing $\sum_i t_i$ and $t_i \geq 0$, this implies

$$t_i = |x_i - y_i|.$$

So, we have our objective:

$$\min \sum_{i=1}^{n} t_i.$$

2.

$$\|x\|_1 + \|y\|_1 \geq 2.$$

Write each variable with its positive and negative parts:

$$x_i = x_i^+ - x_i^-,$$

$$y_i = y_i^+ - y_i^-,$$

where

$$x_i^+, x_i^-, y_i^+, y_i^- \geq 0.$$

Introduce binary variables $s_i, r_i \in \{0, 1\}$ with the conditions:

$$x_i^+ \leq M s_i,$$

$$x_i^- \leq M(1 - s_i),$$

$$y_i^+ \leq M r_i,$$

$$y_i^- \leq M(1 - r_i).$$

This ensures that at most one of the positive and negative parts is nonzero. Therefore,

$$|x_i| = x_i^+ + x_i^-,$$

and

$$|y_i| = y_i^+ + y_i^-.$$

Hence,

$$\|x\|_1 + \|y\|_1 \geq 2$$

becomes

$$\boxed{\sum_{i=1}^{n} \left(x_i^+ + x_i^- + y_i^+ + y_i^-\right) \geq 2.}$$

3.

$$\|y\|_0 \leq 5.$$

Let $z_i \in \{0, 1\}$ and impose

$$\boxed{-M z_i \leq y_i \leq M z_i, \qquad i = 1, \ldots, n,}$$

and

$$\boxed{\sum_{i=1}^{n} z_i \leq 5.}$$

If $z_i = 0$, then $y_i = 0$. Thus, at most five components of y can be nonzero.

4.

$$\|x\|_\infty \leq M$$

is equivalent to

$$\boxed{-M \leq x_i \leq M, \qquad i = 1, \ldots, n,}$$

and

$$\|y\|_\infty \leq M$$

is equivalent to

$$\boxed{-M \leq y_i \leq M, \qquad i = 1, \ldots, n.}$$

All parts of the problem can be represented using linear constraints and/or binary variables, so the problem can be reformulated as a mixed-integer linear optimization problem. □

(d)

False.

Let

$$r = y - X\beta.$$

The inner maximization:

$$\max_{\|\delta\|_2 \leq \lambda} \|r + \delta\|_2.$$

By the triangle inequality,

$$\|r + \delta\|_2 \leq \|r\|_2 + \|\delta\|_2.$$

Since

$$\|\delta\|_2 \leq \lambda,$$

$$\|r + \delta\|_2 \leq \|r\|_2 + \lambda.$$

This upper bound can be achieved by (for $r \neq 0$)

$$\delta = \lambda \frac{r}{\|r\|_2}.$$

This choice uses the full budget $\|\delta\|_2 = \lambda$ and points δ in the same direction as r, so it increases the residual norm as much as possible.

Since δ points in the same direction as r, the triangle inequality can proceed with equality:

$$\begin{aligned}
\|r + \delta\|_2 &= \left\| r + \lambda \frac{r}{\|r\|_2} \right\|_2 \\
&= \left\| \left(1 + \frac{\lambda}{\|r\|_2} \right) r \right\|_2 \\
&= \left(1 + \frac{\lambda}{\|r\|_2} \right) \|r\|_2 \\
&= \|r\|_2 + \lambda.
\end{aligned}$$

Hence,

$$\max_{\|\delta\|_2 \leq \lambda} \|y + \delta - X\beta\|_2 = \|y - X\beta\|_2 + \lambda.$$

The robust optimization problem is

$$\boxed{\min_{\beta} \left(\|y - X\beta\|_2 + \lambda \right)}$$

not

$$\min_{\beta} \left(\|y - X\beta\|_2 + \lambda \|\beta\|_2 \right).$$

□

(e)

False.

The cutting-plane algorithm is not run directly on the big-M formulation. The big-M formulation keeps the regression coefficients β and uses binary variables z_j to determine which coefficients can be nonzero.

Instead, Theorem 3.2 rewrites the problem as

$$\min_{s \in S_k^p} \frac{1}{2} y^T \left(I_n + \gamma \sum_{j \in [p]} s_j K_j \right)^{-1} y,$$

where

$$K_j = X_j X_j^T$$

and $s_j = 1$ if feature j is selected and $s_j = 0$ otherwise.

This removes β from the optimization and does not require big-M constraints.

If we call the objective $c(s)$, the problem is simply

$$\min_{s \in S_k^p} c(s).$$

Since $c(s)$ is convex, the cutting-plane algorithm approximates it using linear cuts

$$\eta \geq c(s_i) + \nabla c(s_i)^T (s - s_i),$$

where s_i is the feature selection found at iteration i. More cuts are added as the algorithm runs, and this gives us a better approximation of $c(s)$.

Therefore, the cutting-plane algorithm is run on the formulation from Theorem 3.2, not directly on the big-M formulation.

□

(f)

True.

Lasso solves

$$\min_{\beta} \|y - X\beta\|_2 + \lambda \|\beta\|_1,$$

where

$$\|\beta\|_1 = \sum_{j=1}^{p} |\beta_j|.$$

The ℓ_1 penalty penalizes the magnitude of the regression coefficients. As λ increases, the coefficients are pushed toward zero, and some may become exactly zero. Therefore, Lasso encourages sparsity and can effectively remove features from the model.

Sparse regression, however, directly imposes a constraint on the number of nonzero coefficients, such as

$$\|\beta\|_0 \leq k.$$

For example, if $k = 2$, sparse regression guarantees that at most two coefficients are nonzero. Lasso may still have more than two nonzero coefficients depending on the value of λ.

Thus, Lasso encourages sparsity through an ℓ_1 penalty, while sparse regression explicitly restricts the number of selected features. The two problems are therefore not equivalent in general.

Question 2

Part A

(a.)

By definition of the induced norm,

$$\|\Delta\|_{(\infty,\infty)} = \max_{x\neq 0} \frac{\|\Delta x\|_\infty}{\|x\|_\infty}.$$

For any $x \neq 0$, define

$$\tilde{x} = \frac{x}{\|x\|_\infty},$$

so that $\|\tilde{x}\|_\infty = 1$. Since $x = \|x\|_\infty \tilde{x}$, we have

$$\frac{\|\Delta x\|_\infty}{\|x\|_\infty} = \frac{\|\Delta(\|x\|_\infty \tilde{x})\|_\infty}{\|x\|_\infty} = \|\Delta\tilde{x}\|_\infty.$$

Thus, it is sufficient to maximize over vectors with $\|x\|_\infty = 1$, giving

$$\|\Delta\|_{(\infty,\infty)} = \max_{\|x\|_\infty=1} \|\Delta x\|_\infty.$$

Let δ_i^T denote each row i of Δ. Then

$$\Delta x = \begin{pmatrix} \delta_1^T x \\ \vdots \\ \delta_n^T x \end{pmatrix},$$

$$\|\Delta x\|_\infty = \max_{i\in[n]} |\delta_i^T x|.$$

$$\|\Delta\|_{(\infty,\infty)} = \max_{\|x\|_\infty=1} \max_{i\in[n]} |\delta_i^T x|.$$

Since both operations are maximizations (hint (i)), their order can be exchanged:

$$\|\Delta\|_{(\infty,\infty)} = \max_{i\in[n]} \max_{\|x\|_\infty=1} |\delta_i^T x|.$$

Moving onto hint (ii.), we need to show upper and lower bound can be established. We now show that, for each row i,

$$\max_{\|x\|_\infty=1} |\delta_i^T x| = \|\delta_i\|_1.$$

For any x satisfying $\|x\|_\infty = 1$, we can expand $\delta_i^T x$ as a dot product:

$$\delta_i^T x = \sum_{j=1}^p \delta_{ij} x_j.$$

We then apply the triangle inequality and take absolute values:

$$|\delta_i^T x| = \left|\sum_{j=1}^p \delta_{ij} x_j\right| \leq \sum_{j=1}^p |\delta_{ij} x_j| = \sum_{j=1}^p |\delta_{ij}||x_j|.$$

Since $\|x\|_\infty = 1$, we have $|x_j| \leq 1$ for every j. Thus,

$$|\delta_i^T x| \leq \sum_{j=1}^p |\delta_{ij}||x_j| \leq \sum_{j=1}^p |\delta_{ij}| = \|\delta_i\|_1.$$

Therefore,

$$\max_{\|x\|_\infty=1} |\delta_i^T x| \le \|\delta_i\|_1.$$

For the reverse inequality lower bound, choose

$$x_j = \operatorname{sign}(\delta_{ij}).$$

This choice satisfies $\|x\|_\infty = 1$, and

$$\delta_i^T x = \sum_{j=1}^{p} \delta_{ij} \operatorname{sign}(\delta_{ij}) = \sum_{j=1}^{p} |\delta_{ij}| = \|\delta_i\|_1.$$

So we have,

$$\max_{\|x\|_\infty=1} |\delta_i^T x| \ge \|\delta_i\|_1.$$

Combine the two inequalities:

$$\max_{\|x\|_\infty=1} |\delta_i^T x| = \|\delta_i\|_1.$$

Substituting this back into the induced norm gives

$$\|\Delta\|_{(\infty,\infty)} = \max_{i\in[n]} \|\delta_i\|_1.$$

□

The switch from 1 to ∞ comes from the dual-norm relationship between ℓ_1 and ℓ_∞. In particular, as shown above,

$$\max_{\|x\|_\infty=1} |\delta_i^T x| = \|\delta_i\|_1.$$

Thus, if we constrain x using the ℓ_∞ norm, we produce the ℓ_1 norm of the row δ_i. This is because ℓ_1 and ℓ_∞ are dual norms, so this explains why an ℓ_1 row budget corresponds to a first induced-norm index of ∞.

If instead the row budget were

$$\|\delta_i\|_2 \le \rho \qquad \text{for all } i,$$

then the first induced-norm index would be 2.

From the definition of the induced norm, we consider

$$\|\Delta\|_{(2,\infty)} = \max_{x\neq 0} \frac{\|\Delta x\|_\infty}{\|x\|_2}.$$

Since scaling x does not change this ratio, we can normalize x by dividing it by $\|x\|_2$. Therefore, it is sufficient to consider vectors satisfying

$$\|x\|_2 = 1.$$

For each row δ_i, by the Cauchy–Schwarz inequality,

$$|\delta_i^T x| \le \|\delta_i\|_2 \|x\|_2.$$

Since $\|x\|_2 = 1$, this gives

$$|\delta_i^T x| \le \|\delta_i\|_2.$$

Therefore,

$$\max_{\|x\|_2=1} |\delta_i^T x| \le \|\delta_i\|_2.$$

To show that this upper bound can be attained, choose

$$x = \frac{\delta_i}{\|\delta_i\|_2}.$$

This is a valid choice because

$$\|x\|_2 = \left\| \frac{\delta_i}{\|\delta_i\|_2} \right\|_2 = \frac{\|\delta_i\|_2}{\|\delta_i\|_2} = 1.$$

$$|\delta_i^T x| = \left| \delta_i^T \frac{\delta_i}{\|\delta_i\|_2} \right| = \frac{\delta_i^T \delta_i}{\|\delta_i\|_2}.$$

Since

$$\delta_i^T \delta_i = \|\delta_i\|_2^2,$$

$$|\delta_i^T x| = \frac{\|\delta_i\|_2^2}{\|\delta_i\|_2} = \|\delta_i\|_2.$$

Thus,

$$\max_{\|x\|_2=1} |\delta_i^T x| \geq \|\delta_i\|_2.$$

Combining the upper and lower bounds gives

$$\max_{\|x\|_2=1} |\delta_i^T x| = \|\delta_i\|_2.$$

Thus, ℓ_2 is self-dual, corresponding to a first induced-norm index of 2.

Therefore, an ℓ_2 constraint on x produces the ℓ_2 norm of each row δ_i, so the first induced-norm index is 2. The second index remains ∞ because $\|\Delta x\|_\infty$ takes the maximum over the rows. Thus, the corresponding uncertainty set is

$$\mathcal{U}_{(2,\infty)}.$$

(b.)

For $n = 1$ and $p = 2$, the perturbation matrix consists of a single row, so each possible perturbation can be represented as a point in the $(\Delta_{11}, \Delta_{12})$ plane.

$$\delta_1 = (\Delta_{11}, \Delta_{12}) \in \mathbb{R}^2.$$

For $\mathcal{A}$, since there is only one row,

$$\|\Delta_j\|_\infty \leq \lambda$$

$$|\Delta_{11}| \leq \lambda, \qquad |\Delta_{12}| \leq \lambda.$$

$$\mathcal{A} = \{(\Delta_{11}, \Delta_{12}) : |\Delta_{11}| \leq \lambda,\ |\Delta_{12}| \leq \lambda\},$$

which is a square centered at the origin with corners $(\pm\lambda, \pm\lambda)$.

For $\mathcal{B}$,

$$\|\delta_1\|_1 \leq \rho,$$

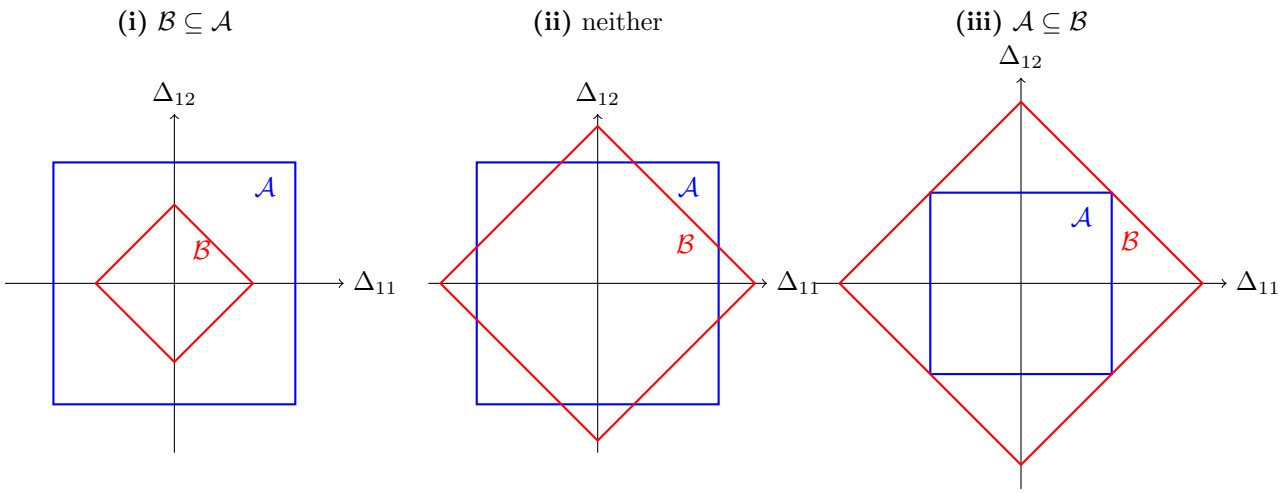
$$|\Delta_{11}| + |\Delta_{12}| \leq \rho.$$

$$\mathcal{B} = \{(\Delta_{11}, \Delta_{12}) : |\Delta_{11}| + |\Delta_{12}| \leq \rho\},$$

which is a diamond centered at the origin with vertices

$$(\rho, 0), \quad (-\rho, 0), \quad (0, \rho), \quad (0, -\rho).$$

The three possible cases are shown below.



For case (i), $\mathcal{B} \subseteq \mathcal{A}$, the diamond must fit entirely inside the square. The vertices of $\mathcal{B}$ extend a distance ρ along each coordinate axis, while $\mathcal{A}$ extends a distance λ. Thus,

$$\boxed{\rho \leq \lambda}.$$

For case (ii), neither set is contained in the other. If $\rho > \lambda$, the vertices $(\rho, 0)$ and $(0, \rho)$ of $\mathcal{B}$ extend outside $\mathcal{A}$, so

$$\mathcal{B} \not\subseteq \mathcal{A}.$$

However, the corner (λ, λ) of $\mathcal{A}$ has ℓ_1 norm

$$|\lambda| + |\lambda| = 2\lambda.$$

If $\rho < 2\lambda$, this corner does not belong to $\mathcal{B}$, so

$$\mathcal{A} \not\subseteq \mathcal{B}.$$

Therefore, neither set contains the other when

$$\boxed{\lambda < \rho < 2\lambda}.$$

For case (iii), $\mathcal{A} \subseteq \mathcal{B}$, the diamond must be large enough to contain all four corners of the square. At a corner, for example (λ, λ),

$$|\Delta_{11}| + |\Delta_{12}| = |\lambda| + |\lambda| = 2\lambda.$$

Thus, all of $\mathcal{A}$ is contained in $\mathcal{B}$ when

$$\boxed{\rho \geq 2\lambda}.$$

(c.)

For $n = 1$ and a general number of features p, the perturbation is

$$\delta_1 = (\Delta_{11}, \ldots, \Delta_{1p}) \in \mathbb{R}^p.$$

The two sets are

$$\mathcal{A} = \{\delta_1 : \|\delta_1\|_\infty \leq \lambda\}, \qquad \mathcal{B} = \{\delta_1 : \|\delta_1\|_1 \leq \rho\}.$$

For case (i), we want $\mathcal{B} \subseteq \mathcal{A}$. If $\delta_1 \in \mathcal{B}$, then

$$\|\delta_1\|_1 \leq \rho.$$

The largest individual coordinate cannot exceed the sum of the absolute values of all coordinates:

$$\|\delta_1\|_\infty \leq \|\delta_1\|_1 \leq \rho.$$

Therefore, if

$$\rho \leq \lambda,$$

then $\|\delta_1\|_\infty \le \lambda$, so $\delta_1 \in \mathcal{A}$. Hence,

$$\boxed{\mathcal{B} \subseteq \mathcal{A} \quad \Longleftrightarrow \quad \rho \le \lambda.}$$

This condition is unchanged as p grows because the total ℓ_1 budget remains ρ, so no individual perturbation can exceed ρ.

For case (iii), we want $\mathcal{A} \subseteq \mathcal{B}$. Since $\mathcal{A}$ allows each of the p coordinates to have magnitude up to λ, the largest possible ℓ_1 norm in $\mathcal{A}$ is

$$\lambda + \cdots + \lambda = p\lambda.$$

For example, this value is attained at the corner

$$\delta_1 = (\lambda, \ldots, \lambda).$$

Therefore, for $\mathcal{B}$ to contain every point in $\mathcal{A}$, its ℓ_1 budget must satisfy

$$\rho \ge p\lambda.$$

Hence,

$$\boxed{\mathcal{A} \subseteq \mathcal{B} \quad \Longleftrightarrow \quad \rho \ge p\lambda.}$$

Unlike case (i), this condition grows with p because each additional feature can contribute up to λ to the total ℓ_1 error.

For completeness, between these two cases,

$$\boxed{\lambda < \rho < p\lambda,}$$

neither set contains the other. In this range, $\mathcal{B}$ can extend outside $\mathcal{A}$ by placing its budget on one coordinate, while the corners $(\pm\lambda, \ldots, \pm\lambda)$ of $\mathcal{A}$ lie outside $\mathcal{B}$. Thus, the "neither" range becomes wider as p grows.

Part B

From Part A, we have

$$\mathcal{A} = \mathcal{U}_{(1,\infty)}$$

with radius λ, and

$$\mathcal{B} = \mathcal{U}_{(\infty,\infty)}$$

with radius ρ.

With the robust optimization result:

$$\max_{\Delta \in \mathcal{A},\ \Delta' \in \mathcal{B}} \|y - (X + \Delta + \Delta')\beta\|_\infty = \|y - X\beta\|_\infty + \lambda\|\beta\|_1 + \rho\|\beta\|_\infty.$$

Therefore, the original problem is equivalent to

$$\min_{\beta} \quad \|y - X\beta\|_\infty + \lambda\|\beta\|_1 + \rho\|\beta\|_\infty$$

subject to

$$\|\beta\|_0 \leq k.$$

1. $\|y - X\beta\|_\infty$

$t \geq 0$ to represent $y_i - X_i\beta$. For each observation $i = 1, \ldots, n$,

$$y_i - X_i\beta \leq t,$$

$$-(y_i - X_i\beta) \leq t.$$

Which imply

$$|y_i - X_i\beta| \leq t \qquad \forall i,$$

Since t is minimized in the objective, it will equal the largest absolute residual:

$$t = \|y - X\beta\|_\infty.$$

2. $\lambda\|\beta\|_1$

$$\|\beta\|_1 = \sum_{j=1}^{p} |\beta_j|,$$

let $u_j \geq 0$ for $j = 1, \ldots, p$ and impose

$$u_j \geq \beta_j,$$

$$u_j \geq -\beta_j.$$

Thus,

$$u_j \geq |\beta_j|.$$

Since the objective minimizes $\sum_{j=1}^{p} u_j$, at optimality

$$u_j = |\beta_j|,$$

$$\sum_{j=1}^{p} u_j = \|\beta\|_1.$$

The ℓ_1 penalty becomes

$$\lambda \sum_{j=1}^{p} u_j.$$

3. $\rho\|\beta\|_\infty$

Let $v \geq 0$ represent the largest absolute coefficient. For each $j = 1, \ldots, p$, impose

$$-v \leq \beta_j \leq v.$$

Therefore,

$$|\beta_j| \leq v \qquad \forall j.$$

Since the objective minimizes v, at optimality

$$v = \max_j |\beta_j| = \|\beta\|_\infty.$$

The ℓ_∞ penalty becomes

$$\rho v.$$

4. $\|\beta\|_0 \leq k$

Introduce binary variables

$$z_j \in \{0, 1\}, \qquad j = 1, \ldots, p,$$

where $z_j = 1$ allows feature j to be selected and $z_j = 0$ forces its coefficient to zero.

Using the given big-M constants M_j, impose

$$-M_j z_j \leq \beta_j \leq M_j z_j \qquad j = 1, \ldots, p.$$

If $z_j = 0$, then $\beta_j = 0$. If $z_j = 1$, then β_j may take any value between $-M_j$ and M_j.

To allow at most k nonzero coefficients, impose

$$\sum_{j=1}^{p} z_j \leq k.$$

Putting all of the above together gives

$$\boxed{\min_{\beta, t, u, v, z} \quad t + \lambda \sum_{j=1}^{p} u_j + \rho v}$$

subject to

$$\begin{aligned}
y_i - X_i\beta &\leq t, \qquad i = 1, \ldots, n, \\
-(y_i - X_i\beta) &\leq t, \qquad i = 1, \ldots, n, \\
u_j &\geq \beta_j, \qquad j = 1, \ldots, p, \\
u_j &\geq -\beta_j, \qquad j = 1, \ldots, p, \\
-v &\leq \beta_j \leq v, \qquad j = 1, \ldots, p, \\
-M_j z_j &\leq \beta_j \leq M_j z_j, \qquad j = 1, \ldots, p, \\
\sum_{j=1}^{p} z_j &\leq k, \\
z_j &\in \{0, 1\}, \qquad j = 1, \ldots, p, \\
t \geq 0, &\qquad u_j \geq 0, \qquad v \geq 0.
\end{aligned}$$

(β_j is the regression coefficient for feature j, t represents $\|y - X\beta\|_\infty$, u_j represents $|\beta_j|$, v represents $\|\beta\|_\infty$, and z_j indicates whether feature j is selected.)

Homework 1 — Question 3

This notebook implements the robust sparse-regression MILP from Question 2, Part B, and reports the requested results for Question 3 using `sublets.csv`.

Imports

In [1]:
```
using JuMP, Gurobi, CSV, DataFrames, LinearAlgebra, Statistics
```

Load and preprocess data

We center `rent` and standardize every feature column (no intercept).

In [2]:
```
df = CSV.read("sublets.csv", DataFrame)
feature_names = names(df)[2:end]   # all columns except rent

y = Vector{Float64}(df.rent)
y = y .- mean(y)

X = Matrix{Float64}(df[:, Not(:rent)])
X = (X .- mean(X, dims=1)) ./ std(X, dims=1)

n, p = size(X)
println("n = $n, p = $p")
println("features: ", feature_names)
```

```
n = 30, p = 7
features:
["sqft", "beds", "baths", "walk_t", "laundry", "parking", "floor"]
```

The MILP solver

Complete the function below. It should implement your model from Question 2 Part B.

Return a `NamedTuple` with fields `obj`, `u`, `w`, `β`, `s`.

In [3]:
```
function solve_robust_sparse(X, y, λ, ρ, k; M=1e3, solver_output=0)
    n, p = size(X)
    @assert length(y) == n

    model = Model(Gurobi.Optimizer)
    set_optimizer_attribute(model, "OutputFlag", solver_output)
```

```
    ################## ADD YOUR CODE AFTER THIS LINE ##################

    @variable(model, β[1:p])            # regression coefficients
    @variable(model, s[1:p], Bin)       # s[j] = 0 forces feature j to
    @variable(model, t[1:p] >= 0)       # t[j] ≥ |β[j]|
    @variable(model, u >= 0)            # u ≥ ‖y − Xβ‖∞
    @variable(model, w >= 0)            # w ≥ ‖β‖∞

    # u ≥ ‖y − Xβ‖∞ :  −u ≤ yᵢ − xᵢᵀβ ≤ u  for every listing i
    resid = y .- X * β
    @constraint(model, [i=1:n],  resid[i] <= u)
    @constraint(model, [i=1:n], -resid[i] <= u)

    # t[j] ≥ |β[j]|
    @constraint(model, [j=1:p],  β[j] <= t[j])
    @constraint(model, [j=1:p], -β[j] <= t[j])

    # w ≥ |β[j]| for every j, i.e. w ≥ ‖β‖∞
    @constraint(model, [j=1:p], t[j] <= w)

    # big-M link: s[j] = 0 ⟹ β[j] = 0
    @constraint(model, [j=1:p],  β[j] <= M * s[j])
    @constraint(model, [j=1:p], -β[j] <= M * s[j])

    # cardinality: ‖β‖₀ ≤ k
    @constraint(model, sum(s) <= k)

    @objective(model, Min, u + λ * sum(t) + ρ * w)

    optimize!(model)

    status = termination_status(model)
    status == MOI.OPTIMAL || error("Gurobi did not solve to optimality

    β_val   = value.(β)
    s_val   = value.(s)
    u_val   = value(u)
    w_val   = value(w)
    obj_val = objective_value(model)

    ################## ADD YOUR CODE BEFORE THIS LINE ##################

    return (obj=obj_val, u=u_val, w=w_val, β=β_val, s=s_val)
end
```

Out[3]: solve_robust_sparse (generic function with 1 method)

In [ ]:
```
"""Pretty-print a solution."""
function report_solution(sol; feature_names=feature_names, atol=1e-6)
    println("objective = $(round(sol.obj, digits=4))")
    println("u         = $(round(sol.u, digits=4))")
```

```
    println("w         = $(round(sol.w, digits=4))")
    selected = findall(j -> sol.s[j] > 0.5 && abs(sol.β[j]) > atol, 1:
    println("selected features:")
    for j in selected
        println("  $(feature_names[j]): β = $(round(sol.β[j], digits=4
    end
    if isempty(selected)
        println("  (none)")
    end
    return selected
end
```

Out[ ]: report_solution

Part (a)

For $\lambda = 0.25$, $\rho = 0.5$, and $k = 3$, the optimal objective value is **404.5056**, with optimal residual level

$$u = \|y - X\beta\|_\infty = \mathbf{196.8252}.$$

The selected features and coefficients are:

- `sqft` : $\beta_{\text{sqft}} = 137.2516$
- `beds` : $\beta_{\text{beds}} = 188.6077$
- `laundry` : $\beta_{\text{laundry}} = 127.6468$

The cell below generates these results.

In [5]:
```
sol_a = solve_robust_sparse(X, y, 0.25, 0.5, 3)
report_solution(sol_a);
```

```
Set parameter Username
Set parameter LicenseID to value 2857457
Academic license - for non-commercial use only - expires 2027-08-26
objective = 404.5056
u         = 196.8252
w         = 188.6077
selected features:
  sqft: β = 137.2516
  beds: β = 188.6077
  laundry: β = 127.6468
```

Part (b)

The cell below compares the three uncertainty regimes with $k = 3$.

In [6]:
```
cases_b = [
```

```
    ("A only (λ=0.25, ρ=0)", 0.25, 0.0),
    ("B only (λ=0, ρ=1)",    0.0,  1.0),
    ("both   (λ=0.25, ρ=0.5)", 0.25, 0.5),
]

for (label, λ, ρ) in cases_b
    println("="^60)
    println(label)
    sol = solve_robust_sparse(X, y, λ, ρ, 3)
    report_solution(sol)
    println()
end
```

```
============================================================
A only (λ=0.25, ρ=0)
Set parameter Username
Set parameter LicenseID to value 2857457
Academic license - for non-commercial use only - expires 2027-08-26
objective = 306.1088
u         = 191.6652
w         = 216.0
selected features:
  sqft: β = 120.4273
  beds: β = 215.1954
  laundry: β = 122.1519

============================================================
B only (λ=0, ρ=1)
Set parameter Username
Set parameter LicenseID to value 2857457
Academic license - for non-commercial use only - expires 2027-08-26
objective = 376.0865
u         = 230.2808
w         = 145.8057
selected features:
  sqft: β = 137.2516
  beds: β = 145.8057
  laundry: β = 145.8057

============================================================
both   (λ=0.25, ρ=0.5)
Set parameter Username
Set parameter LicenseID to value 2857457
Academic license - for non-commercial use only - expires 2027-08-26
objective = 404.5056
u         = 196.8252
w         = 188.6077
selected features:
  sqft: β = 137.2516
  beds: β = 188.6077
  laundry: β = 127.6468
```

Answer to (b)

The selected features and coefficients under each uncertainty regime are:

$\mathcal{A}$ only: $(\lambda, \rho) = (0.25, 0)$

- `sqft` : $\beta_{\text{sqft}} = 120.4273$
- `beds` : $\beta_{\text{beds}} = 215.1954$
- `laundry` : $\beta_{\text{laundry}} = 122.1519$

$\mathcal{B}$ only: $(\lambda, \rho) = (0, 1)$

- `sqft` : $\beta_{\text{sqft}} = 137.2516$
- `beds` : $\beta_{\text{beds}} = 145.8057$
- `laundry` : $\beta_{\text{laundry}} = 145.8057$

Both: $(\lambda, \rho) = (0.25, 0.5)$

- `sqft` : $\beta_{\text{sqft}} = 137.2516$
- `beds` : $\beta_{\text{beds}} = 188.6077$
- `laundry` : $\beta_{\text{laundry}} = 127.6468$

The ℓ_1 penalty from $\mathcal{A}$ charges every unit of total coefficient magnitude, so it shrinks the coefficients overall but does not directly prevent one coefficient from being much larger than the others. The ℓ_∞ penalty from $\mathcal{B}$ charges only the largest magnitude, encouraging the largest coefficients to level off.

Part (c)

The cell below compares the sparse model ($k = 3$) with the dense model ($k = 7$) at $\lambda = 0.25$ and $\rho = 0.5$.

In [7]:
```
println("===== k = 3 (sparse) =====")
sol_sparse = solve_robust_sparse(X, y, 0.25, 0.5, 3)
report_solution(sol_sparse)
println()

println("===== k = 7 (dense) =====")
sol_dense = solve_robust_sparse(X, y, 0.25, 0.5, 7)
report_solution(sol_dense);
```

```
===== k = 3 (sparse) =====
Set parameter Username
Set parameter LicenseID to value 2857457
Academic license - for non-commercial use only - expires 2027-08-26
objective = 404.5056
u         = 196.8252
w         = 188.6077
selected features:
  sqft: β = 137.2516
  beds: β = 188.6077
  laundry: β = 127.6468

===== k = 7 (dense) =====
Set parameter Username
Set parameter LicenseID to value 2857457
Academic license - for non-commercial use only - expires 2027-08-26
objective = 372.8233
u         = 126.5214
w         = 196.3027
selected features:
  sqft: β = 155.9765
  beds: β = 196.3027
  walk_t: β = -60.5236
  laundry: β = 113.3056
  parking: β = 15.6863
  floor: β = -50.8071
```

Answer to (c)

The optimal residual levels are $u = 196.8252$ for the sparse model ($k = 3$) and $u = 126.5214$ for the dense model ($k = 7$).

The nonzero coefficients for $k = 7$ are:

- `sqft` : $\beta_{\text{sqft}} = 155.9765$
- `beds` : $\beta_{\text{beds}} = 196.3027$
- `walk_t` : $\beta_{\text{walk_t}} = -60.5236$
- `laundry` : $\beta_{\text{laundry}} = 113.3056$
- `parking` : $\beta_{\text{parking}} = 15.6863$
- `floor` : $\beta_{\text{floor}} = -50.8071$

By insisting on at most three features, a student gains a simpler, faster-to-use rule but loses worst-case predictive fit, since the residual level increases from 126.5214 to 196.8252.